

Chapter 61 — Urologic Considerations in Pediatric Renal Transplantation

- Children who need **kidney replacement therapy (KRT)** are **more likely than adults to have a congenital anomaly of the kidney and urinary tract (CAKUT)**, including obstructive uropathy.
- The urologist's task is to make sure the condition that destroyed the native kidneys does not destroy the graft. Urology should be part of the multidisciplinary team **before end-stage kidney disease (ESKD)**.
- **Five therapeutic goals:** (1) **unimpeded drainage** from kidney to lower tract · (2) **a capacious, low-pressure reservoir** giving socially acceptable continence · (3) **volitional emptying** · (4) **no infection** · (5) **the fewest operations**.
- **Avoid a reactive approach** (referral only once the graft is in jeopardy); optimise the lower tract early.
- **Chronic rejection is unavoidable, but every added insult accelerates it.**

Concept — The graft is only as safe as the **bladder it drains into**. Paediatric transplant urology is the work of correcting, **before** transplantation, the lower-tract hostility that caused the original renal failure.

PRETRANSPLANT ASSESSMENT

- **North American Pediatric Renal Trials and Collaborative Studies (NAPRTCS):** a large share of children needing KRT have **congenital obstruction, vesicoureteral reflux (VUR) or renal dysplasia, hypoplasia or aplasia**.

	Children and adolescents	Adults
Commonest age at transplant	13–17 years	—
Leading causes	Congenital: hypoplasia, aplasia, dysplasia, then obstructive uropathy. Acquired: focal segmental glomerulosclerosis (FSGS)	Hypertension and diabetes , which rarely cause ESKD in children
By age	Younger: dysplasia or obstructive uropathy, mainly posterior urethral valves (PUVs) in boys . Older: nephrological causes and neuropathic bladder	—

- **Screen every child for urological problems**, although the yield is low when there is a clear nephrological cause.
- **Basic screen:** detailed history + **renal ultrasound** + **uroflow with postvoid residual (PVR)** if toilet-trained. This rules out most abnormalities.
- **Routine voiding cystourethrogram (VCUG) is not necessary.** Consider it with a history of neuropathic bladder, obstructive uropathy, VUR or previous lower-tract surgery, and with **recurrent febrile urinary tract infection (UTI), unexplained worsening hydroureteronephrosis or refractory voiding problems**.

Focused Assessment

- A known urological abnormality → **targeted assessment** directed by the condition and by the state of the bladder and kidneys.
- **VCUG** shows changes with filling and emptying, the urethra, and the presence and grade of reflux. Abnormal bladder dynamics on any test → **urodynamics**.
- **Indications for urodynamics:** neuropathic bladder; outlet obstruction from PUVs or a large ureterocele; and selected **refractory voiding dysfunction, high-grade hydronephrosis or recurrent UTI**.

- **Urodynamics measure** capacity, **compliance**, detrusor activity, sensation, **emptying** and **sphincter function**, to define graft-threatening dynamics and guide therapy.
- **Therapies that follow:** drugs (see table below), **clean intermittent catheterisation (CIC)**, intradetrusor botulinum toxin, reconstruction.
- **Treatment interacts with emptying:** an antimuscarinic for poor compliance can **impair emptying and so create the need for CIC**. A **sensate or difficult urethra** → **continent catheterisable stoma**.
- **Urodynamics are serial, not one-off:** repeat them to track response and change over time.

Figure legend, Fig. 61.1 (algorithm for bladder evaluation before transplant) — The key element is the urological history.

History positive (recurrent UTI, congenital uropathy, voiding complaints, known dysfunction, defunctionalised bladder) → **renal ultrasound (RUS) + VCUG**, then:

- **Severe reflux or hydronephrosis** → urodynamics.
- **Small bladder capacity (<30%)** → anticholinergic and reassess.
- **Incomplete emptying** → behavioural training.
- **Urethral abnormality** → surgical repair.

The outcomes then branch:

- **Inadequate compliance or capacity** → bladder cycling, CIC and anticholinergics, ± **antireflux surgery** → reassess → still inadequate → **bladder augmentation**.
- **Inadequate emptying** → **CIC ± catheterisable stoma**.
- **Adequate compliance, capacity and emptying** → clear for transplant.

History negative → **RUS + PVR**:

- **Normal** → clear for transplant.
- **Hydronephrosis or incomplete emptying** → join the history-positive pathway.

Footnote: an **anuric child with no urological history needs no further testing**. With neurogenic bladder, PUVs or complex history, consider **bladder cycling and anticholinergics**.

KEY POINTS: UROLOGIC EVALUATION AND MANAGEMENT

- The **five goals** (drainage, low-pressure storage, volitional emptying, no infection, fewest procedures).
- **Most children needing KRT have uropathy**; assessment is directed by the condition.
- **Urodynamics** (capacity, compliance, emptying, sphincter) for neuropathic bladder, PUVs, refractory dysfunction, high-grade hydronephrosis or recurrent UTI.

Concept — Assessment is **history-driven and stepwise**: a negative history needs only ultrasound and residual, and every positive finding funnels towards one question, **whether the bladder can store at low pressure and empty**.

PRETRANSPLANT PREPARATION

Detrusor Overactivity, Low Capacity, and Poor Compliance

- The **commonest early bladder abnormalities in ESKD** are low capacity, detrusor overactivity and poor compliance. This is the typical initial picture with PUVs.
- **Bladder dynamics evolve**. The valve bladder moves from overactivity and low capacity → better capacity and less contractility → **insufficient contractility and poor emptying** (“myogenic failure”).

- **Overactivity and poor compliance are the most dangerous patterns**, because **high pressure reaches the kidney even without reflux**. Leaving them untreated puts the graft at risk.
- **Target**: storage up to **expected capacity for age** at **detrusor pressure <40 cm H₂O** (during filling or at leak), plus social continence.
- **Medical treatment first** (antimuscarinics); **augmentation is second line**. Medical treatment often brings in CIC. Some children delay CIC with **double voiding or Valsalva emptying**, and **α-blockers** can aid emptying.
- **Never assume volitional emptying is effective**: prove it with **repeated low PVRs and uroflowmetry**.
- **Oxybutynin is titrated upwards** because of its side effects, watching catheterised volumes, incontinence and hydronephrosis.
- **Adherence to pretransplant CIC is a test of adherence to the transplant regimen**. A family that cannot manage CIC should be assessed for their ability to manage post-transplant care.

AGENTS FOR BLADDER STORAGE AND EMPTYING BEFORE TRANSPLANT

Agent	Class or target	Role
Oxybutynin	Antimuscarinic (non-selective)	First line for overactivity and poor compliance; titrate up against side effects
Solifenacin, darifenacin	More selective antimuscarinics	Better tolerated
Mirabegron	β₃-adrenoceptor agonist	Better tolerated alternative or adjunct
Botulinum toxin	Intradetrusor injection	Better tolerated; refractory overactivity
α-blockers	Outlet (bladder neck)	Aid emptying

- The table's pattern: **everything except the α-blocker targets storage**, and better storage can worsen emptying, which is where CIC comes in.

Capacity

- The trend **away from augmentation** may have gone **too far**. Augment a **small, non-compliant, high-pressure bladder** when maximal medical treatment and CIC still give pressures that are **safe only briefly or regularly exceed 40 cm H₂O**.
- **There is no compelling evidence that augmentation increases complications during or after transplant**.
- **Choice of segment** depends on anatomy, previous surgery and whether a continent catheterisable stoma is wanted.

BLADDER AUGMENTATION SEGMENTS IN THE TRANSPLANT CANDIDATE

Segment	Rationale	Problem	Current status
Stomach	Acid secretion offsets the acidotic state; fewer infections and stones (theoretical)	Haematuria–dysuria syndrome : acid injures the native bladder, especially in anuric pretransplant patients	Popular in the late 1980s–early 1990s; rarely used now
Ileum, sigmoid	Reliable storage	Mucus, stones, infection, metabolic disturbance; malignancy reports, especially after transplant	Most commonly used ; lifelong surveillance

Segment	Rationale	Problem	Current status
Dilated ureter (ureterocystoplasty)	Urothelial lining	Mixed results, depending on ureteric diameter, compliance and VUR	Always explore first; preferable to bowel (no mucus, stones, infections or metabolic disturbance)

- **Preserve the native ureter** for augmentation, a catheterisable channel, or as a target for the graft ureter.
- **At transplant**, the augment patch rarely causes trouble, but **its pedicle position and orientation must be known. Continent stomas are sited medial to the planned transplant incision.**
- **The usual graft site is retroperitoneal through a Gibson incision**, so intraperitoneal dissection is rarely needed. **Infants and small children (<15 kg) may need aorta and vena cava anastomosis and a transperitoneal approach.**

Figure legend, Fig. 61.2 — A right Gibson incision with retroperitoneal dissection in a small child creates room for an adult-size graft with anastomosis to the aorta and vena cava.

Infections

- **Graft pyelonephritis in an immunocompromised host causes major morbidity and can lead to graft loss.**
- The usual cause is **urinary stasis**: inadequate emptying, persistent obstruction or VUR. **UTIs will not improve after transplant**, so they are dealt with beforehand: **double or triple voiding, or CIC.**
- **Distinguish true UTI from colonisation**, which is **almost universal with CIC.**
- **Persistent UTI despite good emptying and CIC technique** → an anatomical abnormality or chronic colonisation.
- **Colonisation** may respond to **long-term antibiotics, intravesical antimicrobials and regular irrigation.**
- **Anatomical causes** such as diverticula or ureteric stumps → targeted imaging and surgery.

The Defunctionalized Bladder

- **Definition**: a bladder that does not cycle, because of **anuria or supravescical diversion**. Its response to physiological post-transplant volumes is hard to predict.

	Neuropathic	Non-neuropathic
Setting	Now rare ; mostly after suboptimal earlier urological care	PUVs, VUR, or anuric children with no urological history (including after bilateral nephrectomy)
Potential	Hard to predict; maximal function takes time to reach	Salvageable for storage and spontaneous voiding
Approach	Same for both : bladder cycling with CIC, assuming CIC may be lifelong in some	
Stoma decision	—	Depends on spontaneous emptying. Sensate children resist urethral CIC , which favours a stoma

Bladder cycling: catheterise and instil saline in **progressively increasing volumes** with a set dwell time, then drain → start volumes empirically from what is tolerated and increase at regular intervals → add medication for capacity and compliance → **target = expected capacity for age** (age-based formulas) → steady gains without pain or leak predict further expansion → **confirm safe pressure, capacity and compliance by urodynamics** → clear for transplant **only when acceptable**, otherwise augment.

- **A renal allograft must never be used to test whether a bladder works.**
- Cycling also tests **the family's ability to perform CIC.**

Decision to augment the bladder.

- **Early proactive management reduces or delays the need to augment.** A trial of regular CIC, **overnight drainage** and maximal medical therapy identifies the few who need it. **Enterocystoplasty** remains one of the most effective ways to provide storage.
- **Broad indication:** inability to protect the upper tracts and to reach **>75% of expected capacity at <40 cm H₂O** while catheterising **no more often than every 3–4 hours**. These thresholds are **somewhat arbitrary**.

	PUV bladder	Neuropathic bladder
Trajectory	Progresses towards myogenic failure	Borderline parameters may deteriorate rapidly under high post-transplant urine output
Timing of augmentation	Postpone as long as possible	Earlier reconstruction justified

- **Enterocystoplasty complications accumulate over a long life**, and immunosuppression worsens infection, healing and malignancy risk, so augmentation is a **lifelong, individualised commitment**.

KEY POINTS: PRETRANSPLANT PREPARATION

- Commonest abnormality: **low-capacity, high-pressure, poorly compliant bladder**.
- **Augmentation does not clearly raise transplant risk; recurrent pyelonephritis does** (graft loss).
- **Pretransplant CIC and bladder cycling** both treat the bladder and test the family.

Concept — Preparation follows one hierarchy: **drugs → CIC → cycling → augmentation**. Each step is tested before the next, because the bladder changes over time and augmentation is irreversible. The PUV and neuropathic bladders sit on **opposite trajectories**, so the same urodynamic numbers argue for delay in one and for early surgery in the other.

Cutaneous Stomas

- A defunctionalised bladder may sit behind an **ileal or colon conduit, vesicostomy, pyelostomy or ureterostomy**. The donor ureter can drain into these.
- **Contemporary preference is definitive bladder optimisation before transplant**. A long-term stoma or transplant ureterostomy is reserved for **severe neurodevelopmental impairment or inability to tolerate CIC**.
- A **functioning ileal loop** can become the **augmentation patch** during refunctionalisation, even in an anuric child. If augmentation proves unnecessary and the child still makes urine, **reimplant the native kidney(s) into the bladder and discard the loop**.

Reconstruction Strategies

- **Factors to weigh:** dialysis type, likely graft site, timing relative to KRT, need for bowel surgery, and the native kidneys.

Dialysis Issues

- **On peritoneal dialysis (PD):** intraperitoneal surgery usually forces a **temporary or permanent switch to haemodialysis**. Surgery can sometimes be done at PD catheter insertion, but **bowel surgery raises the risk of catheter infection and removal**. **Protect the peritoneum**, because PD is often the best temporary KRT in children.
- **Approaching KRT:** dialysis **before** reconstruction improves metabolic state (healing, tolerance of bowel reabsorption), but extensive surgery may **lose the peritoneum for PD**.

- **On haemodialysis:** risks are **line infection with postoperative sepsis** and **arteriovenous fistula injury** from low flow or pressure.

Graft Placement

- **Very small child, possibly intraperitoneal graft** → move mesenteric pedicles **away from the midline**. Such children rarely have earlier reconstruction, and **most grafts fit extraperitoneally**.
- A previous **psoas hitch or extensive ipsilateral reimplant dissection** makes ipsilateral iliac graft placement difficult.
- In difficult cases, a **detailed operative report with a diagram** is of great value to later surgeons.

Timing

- **Major reconstruction is best done before transplant.** Allow **6–12 weeks** to heal after enterocystoplasty or continent diversion.
- **Unilateral native nephrectomy at transplant is reasonable**, especially on the graft side.
- **Avoid concurrent procedures in infants and small children** because of prolonged anaesthesia and haemodynamic instability. **Prophylactic anticoagulation against graft thrombosis** also raises bleeding risk.

Enterocystoplasty

- It is **feasible on dialysis or near ESKD**, much as in any child. The segment and strategy are chosen on functional anatomy: stoma need, native bladder available, bladder neck and continence, and bowel function.
- **An unsalvageable bladder needing continent diversion should not wait for transplant. Exception:** a child transplanted **in the first 2 years of life**, before continence is expected.
- **Metabolic problems are not an absolute contraindication.** Urine reabsorption may **bring dialysis forward** in advanced chronic kidney disease (CKD). In less marginal cases, better bladder dynamics and fewer infections **may delay ESKD**.

Native Nephrectomy

- This is a multidisciplinary, controversial decision. **In general, leave the native kidneys.**

Remove before transplant	Keep (default)
• Hypertension hard to control on several drugs • Profound nephrotic syndrome with malnutrition or hypercoagulability • Massive polyuria • Recurrent upper-tract infection • Large stone burden • High malignancy risk • Very large kidneys (room for graft or PD) • Massive VUR (relative): stasis and infection under immunosuppression, so removal may be safer than reimplantation	• Water excretion on dialysis or after graft failure, so less fluid restriction • Erythropoietin and vitamin D metabolism • Avoids nephrectomy morbidity • Most grafts fit beside small or atrophic kidneys

APPROACH TO NATIVE NEPHRECTOMY BY DIALYSIS STATUS

Situation	Approach	Reason
On PD, small kidneys	Retroperitoneal; bilateral prone without repositioning (optimal)	PD can restart immediately; leaks if the peritoneum is breached
On PD, large kidneys	Posterior open, or laparoscopic with morcellation	Size

Situation	Approach	Reason
On PD, transperitoneal entry	Open or laparoscopic	Needs temporary haemodialysis ; open adhesions can make PD ineffective
Not on PD	Any ; laparoscopic adhesions are unlikely to spoil future PD	A PD catheter can be placed at the same time if PD is imminent
High operative risk	Renal embolisation	Reasonable alternative

Limiting the Risk of Infection

- **Prevent infection proactively**, not only after systemic illness. Targets: **high-grade reflux**, **persistent hydronephrosis** with or without reflux, and **CIC-dependent children** (colonised and UTI-prone).
- **A non-dilated, non-refluxing native kidney is unlikely to be the source of infection.**
- **Circumcision** in boys at risk of ESKD, especially with **valves**, should be discussed early. It may slow progression to CKD and reduce post-transplant UTI.

Preserving the Ureter(s)

Keep the ureter	Remove the ureter
Normal ureter : transect as high as possible to limit iliac fibrosis. It is a reserve for primary or salvage ureteroureterostomy if the graft's distal ureter narrows. Also keep it if CIC is likely, for a continent stoma (best made before transplant)	High-grade reflux or obstruction with a dilated ureter → total removal to limit infection

Combining Nephrectomy and Transplantation

- **Generally avoided** to limit morbidity and anaesthetic time. It is mostly done to **make room** for an **adult graft in a small child** or beside very large kidneys. The best case is a **single ipsilateral nephrectomy** through the transplant incision.

KEY POINTS: PRETRANSPLANT RECONSTRUCTION

- **Spontaneous emptying** decides the need for a catheterisable stoma.
- **Augment** for capacity **<75%**, pressure **>40 cm H₂O**, CIC **every ≤3 hours**, despite maximal medical therapy.
- **PD** → **intraperitoneal surgery** needs **temporary haemodialysis**; reconstruct **well before** transplant.
- **Native nephrectomy indications** as in the table above.

Concept — Every reconstructive decision is judged by **what it costs the future transplant**: the peritoneum for PD, the iliac fossa for the graft, the ureter for salvage, the native kidney for fluid and hormones. **Preserve by default, and remove only for a named indication.**

TRANSPLANTATION

- The vascular technique is covered in Chapter 92, with the same principles in children. For the paediatric urologist, **lower-tract reconstruction is the intraoperative focus** and the main source of urological complications.

Ureteral Anastomosis

Surgical Techniques and Options

- As in VUR surgery, there are **intravesical** and **extravesical** options. The authors favour a **non-refluxing anastomosis** into the bladder: it is **simple, effective and low-morbidity**, although an antireflux mechanism is not always essential.

Extravesical ureteroneocystostomy (preferred): after reperfusion and haemostasis, **partially fill the bladder** (saline or dilute antibiotic) → clear the anterolateral wall; traction sutures lift the lateral bladder and tension the wall → check the ureter will reach → **detrusorotomy 3–4 cm** down to mucosa, in an anatomical line that avoids kinking the donor ureter → raise detrusor flaps → **excise a small mucosal disc** at the distal trough → trim the ureter and **spatulate 4–5 mm** → **interrupted mucosa-to-mucosa anastomosis** with fine absorbable sutures → **anchor stitches through detrusor and ureteric adventitia** against eversion → close the flaps over the ureter with interrupted absorbable sutures, **not too tight at the entry point**.

TRANSPLANT URETER DRAINAGE OPTIONS

Technique	How	Best suited to
Extravesical ureteroneocystostomy	3–4-cm detrusorotomy; mucosa-to-mucosa	Default
Barry technique	4-cm tunnel between two parallel detrusor incisions ; ureter passed through; shorter tunnels acceptable in children	Alternative non-refluxing route
Ureteroureterostomy (graft to native ureter)	Side-to-side, end-to-side or end-to-end; the proximal native ureter can be ligated with the native kidney left in place (few complications, few later nephrectomies)	Normal non-refluxing native ureter, small bladder, short donor ureter, hostile pelvis ; always the salvage option for stenosis after ureteroneocystostomy

- Whether ureteroureterostomy is superior to ureteroneocystostomy is **unclear**.

Ureteral Stenting

- Routine stenting is debated**, and there is no conclusive evidence for it in children.
- Stent when:** the reimplant is **difficult** (especially into an abnormal bladder), or the graft ureter looks **traumatised or devascularised**.
- In adult series with **deceased-donor grafts**, stenting **reduced ureteric complications at the cost of more infection**.
- Standard:** a short **double-J removed at 4–6 weeks** by cystoscopy, timed with removal of a tunnelled line or PD catheter.
- Early removal at 5–7 days** (stent tied to the urethral catheter tip, or an externalised stent or splint) **may reduce stent complications and infection**.

KEY POINTS: TRANSPLANT URETERONEOCYSTOSTOMY

- Extravesical non-refluxing reimplant preferred**; ureteroureterostomy effective.
- No data for routine stenting**; **early removal** is safe and may cut infection.

COMPLICATIONS

- Vascular and rejection complications are covered in Chapter 92. **Urological complications are uncommon and rarely cost the graft**.

- **Thrombosis is one of the commonest causes of early graft loss in children**, so some centres anticoagulate early, which demands careful haemostasis.
- **Pre-existing obstructive or reflux uropathy increases all urological complications**. The link between PUVs and graft dysfunction is **inconsistent across series**. Apply **special vigilance and a lower threshold to intervene**.

FREQUENCY OF UROLOGICAL COMPLICATIONS: KEY POINTS VERSUS BODY TEXT

Complication	Key Points box	Body text
Ureteric stenosis or obstruction	6%	Hydronephrosis indicates obstruction in up to 8%
Urine leak	3%	<5%
Stones	2%	Up to 5% in one paediatric series, probably <1%; about 2% at 3 years in all-age recipients
Clinically significant reflux	2%	—

- The columns differ because they come from **different series and denominators** (pooled point estimates versus single-series ranges). **Stenosis is commonest; the rest are low single figures**.

Urine Leaks

- **Rare and early**: wound drainage, rising drain output or a perinephric collection. **Fluid creatinine** distinguishes urine from lymph or peritoneal fluid.

Sequence: check every drain, **above all the bladder catheter** (irrigate it; replace it if removed) → **transplant ultrasound**: hydronephrosis means distal obstruction, so consider **percutaneous nephrostomy** → confirm stent position → localise the leak with a **mercaptoacetyltriglycine (MAG3) scan** (if function is adequate) or a **cystogram** (anastomotic leak, or pelvicalyceal leak if a stent allows reflux) → small leak: **observe** → small child with high output: **a larger catheter** may resolve it → persistent or large leak: **percutaneous drainage and/or exploration**.

- **Prevention**: at transplant, place the largest catheter the urethra will take.
- **Repair by cause**:
 - **Bladder-side leak** → revise the anastomosis and drain.
 - **Short distal ureteric necrosis** → mobilise the bladder and reimplant.
 - **Long necrosis** → **ipsilateral or even contralateral native ureter**; otherwise **psoas hitch or bladder flap**. Stent and drain carefully.

Figure legend, Fig. 61.20 — A leak at a **ureteroureterostomy** shown on cystogram: contrast tracks around the stent, refluxes up the native ureter, and a drain is in place. It was first detected as **high drain output with a high creatinine**.

Infection

- **Early** infections relate to the **catheter, stent and instrumentation**. **Late** infection is a relatively frequent, delayed complication that **reflects bladder function** and the original uropathy in an immunosuppressed host.
- **High-grade or worsening hydronephrosis** → pyelonephritis and functional loss, so evaluate promptly. Routine assessment covers emptying, hydronephrosis and **selective VCUG**.
- **On CIC with no correctable cause** → prophylactic antibiotics and bladder irrigation, plus proactive bladder management.

- **Graft VUR with recurrent febrile UTI → surgical correction. Reflux without infection and with normal bladder function → observe.**

Reflux

	Transplant VUR	Native VUR
Ureter	Reimplanted donor ureter	Native UVJ
Cost of pyelonephritis	Greater: a solitary graft in an immunosuppressed host	Lower
Endoscopic injection	Poor resolution (about one-third); higher obstruction risk than for other indications; selected low-grade only	Established option

- **Routine post-transplant screening for reflux is controversial;** some centres do it for risk assessment. A febrile UTI → cystogram.
- **Recurrent febrile UTI with optimised bladder dynamics plus transplant reflux → open or robotic reimplantation.** Uncorrected, the risk of further graft pyelonephritis and **graft loss** is significant. Coexisting bladder dysfunction must be sought, though it may not be correctable.
- **Asymptomatic reflux** (no infection, no bladder dysfunction) → **observe**. Surgery for transplant reflux is **never routine**.
- **Technique:** few comparative data. The authors prefer **intravesical repair, often with extravesical mobilisation**. Use **transtrigonal** if the contralateral native ureter can be avoided; otherwise an **advancement or Politano–Leadbetter** repair. **Stent**.
- **Graft ureter joined to the ipsilateral native ureter** → same options, but injection and dissection go **better** because the patient's own UVJ is preserved.

Figure legend, Fig. 61.22 — Graft glomerular filtration rate (GFR) at 1 and 4 years is lower with ≥ 2 UTIs than with ≤ 1 UTI, and the gap widens by 4 years (Herthelius and Oborn, 2007).

Concept — Transplant reflux is managed by **consequence, not by grade**. Infection plus a solitary immunosuppressed kidney justifies surgery, its absence justifies observation, and **the bladder must be optimised before the ureter is blamed**.

Hydronephrosis and Obstruction

- **Hydronephrosis is frequent** after transplant and signals obstruction in **up to 8%**. The graft is **unusually sensitive to obstruction**, and the **grade of hydronephrosis does not track the degree of impairment**.
- **Rising creatinine and cystatin C** often herald obstruction. **More than half of obstructions occur within 100 days**, so **long-term ultrasound surveillance** is needed.
- **Normal pre-transplant bladder** → **this pattern is ureteric obstruction until proven otherwise**. A known or suspected bladder problem → evaluate and treat both. **Obstruction is commoner with bladder abnormality, especially PUVs**.
- **Isolated VUR is unlikely to cause progressive hydronephrosis and falling function**.

RIISING CREATININE WITH HYDRONEPHROSIS: THE STEPWISE DISCRIMINATING SEQUENCE

Scenario	First step	Next step
Bladder dysfunction suspected	Indwelling catheter (continuous drainage 1–2 weeks, recheck creatinine)	Improves → bladder is the cause ; no change → stent plus catheter, recheck
Mild hydronephrosis + other signs of rejection	Biopsy	Treat rejection medically if the graft is not failing fast; stent if no improvement

Scenario	First step	Next step
No sign of rejection; no improvement with catheter	Decompress: ureteric stent or nephrostomy ($\pm$ biopsy)	Failed retrograde stent → percutaneous drainage → antegrade study and stent

- **Obstruction and rejection can coexist.** Diagnostic tests for obstruction in a graft are **unreliable**, so keep a **low threshold for diagnostic stenting**. Occasionally, an acutely failing graft recovers after stenting even without rejection.
- **Site and cause:** usually **distal stenosis at the reimplant**, often with **ischaemia of the distal donor ureter**. The implant technique and stenting are **not conclusively linked** to obstruction. **PUV bladders are a risk factor**.
- **Extrinsic causes:** **lymphocele** and **post-transplant lymphoproliferative disease (PTLD)**.
- **Treatment:**
 - **Focal narrowing** → **balloon dilatation and stenting for 4–6 weeks**.
 - **Long-term stenting** (adult practice) is of uncertain value in children.
 - **Open repair should not be delayed excessively:** native ureter, bladder flaps, or **pyeloureteral anastomosis** to a healthy retained native ureter.
 - **Avoid non-urothelial substitutes** such as the appendix where possible, and follow reoperative principles (well-vascularised tissue).

Figure legends, Figs 61.23–61.25 — **Obstruction-free survival is significantly lower with a history of PUVs** (Smith et al., 2010). On ultrasound, a **septated lymphocele** compresses the graft ureter and causes hydronephrosis. PTLD appears as an intrarenal mass with lymphadenopathy, and **an allograft mass is PTLD until proven otherwise**.

Bladder Dysfunction

- It **raises infection risk** and can **impair drainage**, which harms the graft. It is hard to separate from ureteric obstruction, so it needs **sequential diagnostic drainage**:
 - **Catheter for 1–2 weeks** → creatinine falls and hydronephrosis improves → **bladder is the cause**.
 - If not → **stent plus catheter** and recheck.
- **Treatment:** behavioural and medical measures and a new or intensified CIC programme. **Augmentation rarely, and only after maximal medical treatment**. A **continent stoma** if urethral CIC is difficult.

Stones

- **Uncommon** (rates in the table above); risk is higher with a **history of stones**.
- With proper management, stones **do not increase graft loss** but are linked to **more UTIs**.
- **Bladder stones** reported in the past were often on **retained suture material**.
- **Donor-derived stones:** **bench flexible ureteroscopy** can clear small stones during cold preparation. Adult data suggest “**donor-gifted**” lithiasis is **rarely troublesome and can be observed**; there are no paediatric data.
- **New asymptomatic stone** → monitor closely and look for a cause. **Treat promptly** if **>1 cm and growing**, or causing obstruction, infection or graft dysfunction.
- Any standard modality may be used, but **choose the one most likely to succeed in a single intervention**, even if it is more complex. Account for graft position, stone size and site, and retrograde access to the ureter.

KEY POINTS: COMPLICATIONS

- **Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.**
- **Leaks early; infection late**, reflecting the bladder; transplant VUR costs more than native VUR.
- **Creatinine rise plus hydronephrosis**: obstruction and rejection may coexist; obstruction is usually distal.
- **Bladder dysfunction** causes infection and can itself obstruct.

Concept — Post-transplant urological complications sort by time: **leaks early, obstruction within about 100 days, infection late**. Nearly all are **amplified by the original bladder**. That is why every diagnostic pathway begins with a catheter, which separates bladder from ureter, before anything more invasive.

SUMMARY

- Success rests on a **clear pretransplant bladder strategy, urologist involvement in every case, diligent surveillance** before damage is irreversible, and **multidisciplinary collaboration**.